Math 190-1: Mathematics of Sustainability **Spring 2026**

Lecture 1: Matrix Review

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1.1 Matrices, Matrix operations, and Solving Linear Systems

Definition 1.1 (Matrix) *A* matrix *is a rectangular array of numbers with* dimension*, or the number of both* rows *and* columns.

Example 1.2 *Let*

$$\mathbf{A} = \begin{pmatrix} -2 & 5 & 6 \\ 5 & 2 & 7 \end{pmatrix} \quad \mathbf{B} = \begin{bmatrix} -8 & -4 \\ 23 & 12 \\ 18 & 10 \end{bmatrix}$$

We say that $\mathbf{A}$ *is a* 2×3 *matrix, and* $\mathbf{B}$ *is a* 3×2 *matrix.*

Remark 1.3 *We always call a matrices' dimensions in* row *by* column *order.*

Definition 1.4 *A* matrix element *is a number in a matrix. It is identified by its* row *by* column *order.*

Example 1.5 *Consider the matrix* $\mathbf{G}$ *where*

$$\mathbf{G} = \begin{pmatrix} 4 & 14 & -7 \\ 18 & 5 & 13 \\ -20 & 4 & 22 \end{pmatrix}$$

The element $g_{21} = 18$*. In general, the element in row* i *and column* j *of matrix* $\mathbf{A}$ *is denoted as* a_{ij}.

1.1.1 Matrices and linear system

Sometimes you have a system of equations that you would want to solve:

$$\begin{aligned} 2x + 5y &= 10 \\ 3x + 4y &= 24 \end{aligned}$$

This can be rewritten as the matrix $\mathbf{A}$:

$$\mathbf{A} = \left(\begin{array}{cc|c} 2 & 5 & 10 \\ 3 & 4 & 24 \end{array}\right).$$

The first two columns represent the x and y. The column after the $\big|$ are the constants. This is called the *augmented matrix*.

1.1.2 Matrix Row Operations

Switch rows

$$\begin{pmatrix} 2 & 5 & 3 \\ 3 & 4 & 6 \end{pmatrix} \to \begin{pmatrix} 2 & 5 & 3 \\ 3 & 4 & 6 \end{pmatrix}$$

We will use the notation $R_1 \leftrightarrow R_2$ to say that row 1 and row 2 are changed.

Multiply a row by a nonzero constant

$$\begin{pmatrix} 2 & 5 & 3 \\ 3 & 4 & 6 \end{pmatrix} \to \begin{pmatrix} 3 \cdot 2 & 3 \cdot 5 & 3 \cdot 3 \\ 3 & 4 & 6 \end{pmatrix}$$

We will use the notation $3R_1 \to R_1$ to say that row 1 and becomes 3 times itself are changed.

Add one row to another

$$\begin{pmatrix} 2 & 5 & 3 \\ 3 & 4 & 6 \end{pmatrix} \to \begin{pmatrix} 2 & 5 & 3 \\ 3+2 & 4+5 & 6+3 \end{pmatrix}$$

We will use the notation $R_1 + R_2 \to R_2$ to say that row 2 becomes row 1 + row 2.

1.1.3 Reduced Row-echelon form & Gaussian elimination

We will now combine the previous two sections to solve problems that we care about. When we are given a system of linear equations (which will come up quite frequently in this class), we would like to solve them. We do this by first putting the system into an augmented matrix and then performing these matrix row operations until we get to a solution.

Definition 1.6 *This process of performing elementary row operations is called* Gaussian elimination*, which leads to a solution which can be figured out when the solution is in* reduced row echelon form*.*

Basically, for each column you zero everything else below out.

Example 1.7 *Solve the following linear system:*

$$\begin{aligned} x + y + z &= 3 \\ x + 2y + 3z &= 0 \\ x + 3y + 4z &= -2 \end{aligned}$$

We first start by constructing the augmented matrix

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 1 & 2 & 3 & 0 \\ 1 & 3 & 4 & -2 \end{array}\right).$$

Then we need to do the elementary row operations to zero out elements below the first 1. The 1 in the first row and column (a_{11}) is special because it is the first nonzero element in that row. We call this a pivot. Reduced Row Echelon form requires that after all is said and done, that each pivot is 1 and the only nonzero in its row.

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 1 & 2 & 3 & 0 \\ 1 & 3 & 4 & -2 \end{array}\right) \overset{R_2 - R_1 \to R_2}{=} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & 2 & -3 \\ 1 & 3 & 4 & -2 \end{array}\right) \overset{R_3 - R_1 \to R_3}{=} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & 2 & -3 \\ 0 & 2 & 3 & -5 \end{array}\right)$$

We have zeroed out the first column, so now we move on to the next column. The next pivot, is the first nonzero element in the second row. Starting with this pivot, we zero everything else out in that row.

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & 2 & -3 \\ 0 & 2 & 3 & -5 \end{array}\right) \overset{R_1 - R_2 \to R_1}{=} \left(\begin{array}{ccc|c} 1 & 0 & -1 & 6 \\ 0 & 1 & 2 & -3 \\ 0 & 2 & 3 & -5 \end{array}\right) \overset{-R_3 + 2R_2 \to R_3}{=} \left(\begin{array}{ccc|c} 1 & 0 & -1 & 6 \\ 0 & 1 & 2 & -3 \\ 0 & 0 & 1 & -1 \end{array}\right)$$

Lastly, we need to zero out everything in the same column as our last pivot.

$$\left(\begin{array}{ccc|c} 1 & 0 & -1 & 6 \\ 0 & 1 & 2 & -3 \\ 0 & 0 & 1 & -1 \end{array}\right) \overset{R_2 - 2R_3 \to R_2}{=} \left(\begin{array}{ccc|c} 1 & 0 & -1 & 6 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{array}\right) \overset{R_1 + R_3 \to R_1}{=} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{array}\right)$$

We now have the matrix in RREF, as there are pivots and for each pivot, it is the only element in each column. We now go from the augmented matrix back to a system of equations, so we have

$$\begin{aligned} x + 0y + 0z &= 5 \\ 0x + y + 0z &= -1 \\ 0x + 0y + z &= -1, \end{aligned}$$

or $x = 5, y = z = -1$, which is the solution of the system of equations. Lastly because our coefficient matrix, the matrix before the $|$ is square, meaning it has 3 rows and three columns (meaning there are three equations with three unknowns x, y, z), RREF becomes the identity matrix, meaning 1's along the diagonal, and 0's everywhere else.

1.1.4 Matrix Addition and Subtraction

Remember that a matrix has dimensions, (rows by columns!), and as long as two matrices have the same number of rows and the same number of columns, they can be added.

Example 1.8 *Let*

$$\mathbf{A} = \begin{pmatrix} 4 & 8 \\ 3 & 7 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 1 & 0 \\ 5 & 2 \end{pmatrix}.$$

To find $\mathbf{A}+\mathbf{B}$ we need to make sure that they have the same dimensions. Since both are 2 rows by 2 columns, we can add them! So

$$\mathbf{A} + \mathbf{B} = \begin{pmatrix} 4 & 8 \\ 3 & 7 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 5 & 2 \end{pmatrix} = \begin{pmatrix} 4+1 & 8+0 \\ 3+5 & 7+2 \end{pmatrix} = \begin{pmatrix} 5 & 8 \\ 8 & 9 \end{pmatrix}$$

We can also do subtraction of two matrices the same way. First, the dimensions have to match up, but if they do, then we subtract each element with its corresponding element in the other matrix.

Example 1.9 *Let*

$$\mathbf{C} = \begin{pmatrix} 2 & 8 \\ 0 & 9 \end{pmatrix}, \mathbf{D} = \begin{pmatrix} 5 & 6 \\ 11 & 3 \end{pmatrix}.$$

To find $\mathbf{C}-\mathbf{D}$ *we need to make sure that they have the same dimensions. Since both are 2 rows by 2 columns, we can add them! So*

$$\mathbf{C}-\mathbf{D} = \begin{pmatrix} 2 & 8 \\ 0 & 9 \end{pmatrix} - \begin{pmatrix} 5 & 6 \\ 11 & 3 \end{pmatrix} = \begin{pmatrix} 2-5 & 8-6 \\ 0-11 & 9-3 \end{pmatrix} = \begin{pmatrix} -3 & 2 \\ -11 & 6 \end{pmatrix}$$

1.1.5 Scalar multiplication

A scalar is just a real number, so scalar multiplication is when the scalar is multiplying an entire matrix.

Example 1.10 *Like before, let*

$$\mathbf{A} = \begin{pmatrix} 4 & 8 \\ 3 & 7 \end{pmatrix},$$

then we could view $\mathbf{A}+\mathbf{A}+\mathbf{A}$ *as*

$$\mathbf{A}+\mathbf{A}+\mathbf{A} = \begin{pmatrix} 4 & 8 \\ 3 & 7 \end{pmatrix} + \begin{pmatrix} 4 & 8 \\ 3 & 7 \end{pmatrix} + \begin{pmatrix} 4 & 8 \\ 3 & 7 \end{pmatrix} = \begin{pmatrix} 4+4+4 & 8+8+8 \\ 3+3+3 & 7+7+7 \end{pmatrix} = \begin{pmatrix} 12 & 24 \\ 9 & 21 \end{pmatrix} = 3\mathbf{A}$$

The last step might have confused you, but let's break it down.

$$3\mathbf{A} = 3 \cdot \begin{pmatrix} 4 & 8 \\ 3 & 7 \end{pmatrix} = \begin{pmatrix} 3\cdot 4 & 3\cdot 8 \\ 3\cdot 3 & 3\cdot 7 \end{pmatrix} = \begin{pmatrix} 12 & 24 \\ 9 & 21 \end{pmatrix}$$

1.2 Matrix Multiplication

This is the first unintuitive definition, but it is the crux of what we will be doing in this class. Let $\mathbf{C} = \mathbf{AB}$, where

$$\mathbf{A} = \begin{pmatrix} 1 & 7 \\ 2 & 4 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 3 & 3 \\ 5 & 2 \end{pmatrix}.$$

We will start by finding the first element of $\mathbf{C}, c_{11}$. This consists of looking at the first row of $\mathbf{A}$ and the first column of $\mathbf{B}$.

$$c_{11} = 1 \cdot 3 + 7 \cdot 5 = 3 + 35 = 38.$$

Now we will try to find $c_{1,2}$, which involves the first row of $\mathbf{A}$ but the <u>second</u> row of $\mathbf{B}$.

$$c_{12} = 1 \cdot 3 + 7 \cdot 2 = 3 + 14 = 17.$$

To finish we see that $c_{21} = 2 \cdot 3 + 4 \cdot 5 = 6 + 20 = 26$ and $c_{22} = 2 \cdot 3 + 4 \cdot 2 = 6 + 8 = 14$. Putting this together, we end up with

$$\mathbf{C} = \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix} = \begin{pmatrix} 38 & 17 \\ 26 & 14 \end{pmatrix}$$

In general, c_{ij} will consist of operating on the i^{th} row of $\mathbf{A}$ and the j^{th} column of $\mathbf{B}$. This means that the number of elements in the i^{th} row of $\mathbf{A}$ (which means the number of columns of $\mathbf{A}$) must be the same as the number of elements in the j^{th} column of $\mathbf{B}$ (which means the number of rows of $\mathbf{B}$).

So for example, we cannot multiply

$$\begin{pmatrix} 2 & 8 & 3 \\ 5 & 4 & 1 \end{pmatrix} \cdot \begin{pmatrix} 4 & 6 & 2 \\ 1 & 3 & 4 \end{pmatrix}.$$

We could start by trying to find the first element. $2 \cdot 4 + 8 \cdot 1 + 3 \cdot ???$. There is nothing there! We could also see that the first matrix is 2×3 and the second matrix is 2×3. We need the number of columns of the first matrix to match the number or rows in the second matrix, i.e. the middle two numbers much match.

But what if we transpose the second matrix, where

$$\begin{pmatrix} 4 & 6 & 2 \\ 1 & 3 & 4 \end{pmatrix}^\top = \begin{pmatrix} 4 & 1 \\ 6 & 3 \\ 2 & 4 \end{pmatrix}.$$

Now that the matrix has 3 rows and 2 columns, we can multiply a 2×3 with a 3×2, to get

$$\begin{pmatrix} 2 & 8 & 3 \\ 5 & 4 & 1 \end{pmatrix} \cdot \begin{pmatrix} 4 & 1 \\ 6 & 3 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 2 \cdot 4 + 8 \cdot 6 + 3 \cdot 2 & 2 \cdot 1 + 8 \cdot 3 + 3 \cdot 4 \\ 5 \cdot 4 + 4 \cdot 6 + 1 \cdot 2 & 5 \cdot 1 + 4 \cdot 3 + 1 \cdot 4 \end{pmatrix} = \begin{pmatrix} 8 + 48 + 6 & 2 + 24 + 12 \\ 20 + 24 + 2 & 5 + 12 + 4 \end{pmatrix} = \begin{pmatrix} 62 & 38 \\ 46 & 21 \end{pmatrix}$$

Notice that this matrix is 2×2. We could have also figured that out by seeing that we are multiplying a 2×3 with a 3×2 matrix. We know it is legal since $3 = 3$, but we can tell the dimensions of the end result by looking at the first dimension of the first matrix and the last dimension of the second matrix, which are both 2, so the final product would be a 2×2 matrix.

In general if we are trying to multiply a matrix with dimensions $n \times m$ and another that is $m \times k$, we see that $m = m$, so multiplication is defined, but the dimensions of the final matrix is going to be $n \times k$.

1.2.1 Determinants and inverses of 2×2 matrices

Now we turn our attention to a different property of matrices, which has connections to much of linear algebra, that we won't have time to delve into. This properties have deep ties to the elementary row operations, rref we already discussed and inverses, which we will talk about next.

Definition 1.11 *A* determinant *is a function that takes square matrices to a real number, where the following properties are*

1. *doing a row replacement on* $\mathbf{A}$ *doesn't change* $\det(\mathbf{A})$.
2. *scaling a row of* $\mathbf{A}$ *by scalar* c*, multiplies the determinant by* c.
3. *swapping two rows of a matrix multiplies the determinant by* -1.
4. *the determinant of the identity matrix* $\mathbf{I}_n$ *is 1.*

Example 1.12 *Using elementary row operations, we are able to determine* $\det\begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}$.

$$\begin{aligned}
\begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix} &\overset{R_1 \leftrightarrow R_2}{=} \begin{pmatrix} 1 & 4 \\ 2 & 1 \end{pmatrix} \\
&\overset{R_2 - 2R_1 \to R_2}{=} \begin{pmatrix} 1 & 4 \\ 0 & -7 \end{pmatrix} \\
&\overset{-1/7R_2 \to R_2}{=} \begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix} \\
&\overset{R_1 - 4R_2 \to R_1}{=} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}
\end{aligned}$$

Now that we have the $\mathbf{I}_2$, *we know the last step means* $\det(\mathbf{I}_2) = 1$. *We now retrace our steps. In the penultimate step, we did a row replacement, which doesn't alter the determinant, so* $\det\begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix} = 1$. *Before that, we divided by* -7, *so the determinant needs to be multiplied by* -7 *to "undo" what we did. This means that* $\det\begin{pmatrix} 1 & 4 \\ 0 & -7 \end{pmatrix} = -7$. *Before that, we did a row replacement, so the determinant is unchanged, i.e. We then did a row swap, so that negates the determinant, which means* $\det\begin{pmatrix} 1 & 4 \\ 2 & 1 \end{pmatrix} = -7$. *Lastly, we did a row swap, which multiplies the determinant by* -1, *meaning the determinant of the original matrix,* $\det\begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix} = 7$.

Since this can be an in depth computation, we can find a shortcut for the 2×2 matrix case.

Definition 1.13 *Let* $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ *be a* 2×2 *matrix. The* determinant *of a* $\mathbf{A}$, *written either as* $\det(\mathbf{A})$ *or*

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ab - bc$$

Example 1.14 *Let* $\mathbf{A} = \begin{pmatrix} 5 & 3 \\ -1 & 4 \end{pmatrix}$ *be a* 2×2 *matrix. The* determinant *of a* $\mathbf{A}$, *written either as* $\det(\mathbf{A})$ *or*

$$\begin{vmatrix} 5 & 3 \\ -1 & 4 \end{vmatrix} = 5 \cdot 4 - 3 \cdot (-1) = 20 - (-3) = 23$$

Definition 1.15 *Let* $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ *be a* 2×2 *matrix. The inverse of a* $\mathbf{A}$ *is*

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Definition 1.16 *Let* $\mathbf{A}$ *be a* $n \times n$ *matrix. We define the* inverse *of* $\mathbf{A}$ *as* $\mathbf{A}^{-1}$ *such that*

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_n$$

Example 1.17 *Let* $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ -3 & 4 \end{pmatrix}$. *We want to find the inverse* $\mathbf{A}^{-1}$. *To do this, we need to start by finding the* $\det(\mathbf{A}) = 1 \cdot 4 - 2(-3) = 4 + 6 = 10$. *Now we switch* a_{11} *and* a_{22} *and negate* a_{12}, a_{21} *but keep them in place, so*

$$\mathbf{A}^{-1} = \frac{1}{10} \begin{pmatrix} 4 & -2 \\ 3 & 1 \end{pmatrix}.$$

Now we check that this is actually an inverse

$$\mathbf{A}\mathbf{A}^{-1} = \frac{1}{10} \begin{pmatrix} 1 & 2 \\ -3 & 4 \end{pmatrix} \begin{pmatrix} 4 & -2 \\ 3 & 1 \end{pmatrix} = \frac{1}{10} \begin{pmatrix} 1 \cdot 4 + 2 \cdot 3 & 1 \cdot (-2) + 2 \cdot 1 \\ -3 \cdot 4 + 4 \cdot 3 & -3 \cdot (-2) + 4 \cdot 1 \end{pmatrix} = \frac{1}{10} \begin{pmatrix} 4 + 6 & 0 \\ 0 & 6 + 4 \end{pmatrix} = \frac{1}{10} \begin{pmatrix} 10 & 0 \\ 0 & 10 \end{pmatrix}$$

Example 1.18 *Let* $\mathbf{A} = \begin{pmatrix} 1 & -2 \\ -3 & 6 \end{pmatrix}$. *We want to find the inverse* $\mathbf{A}^{-1}$. *To do this, we need to start by finding the* $\det(\mathbf{A}) = 1 \cdot 6 - (-2)(-3) = 6 - 6 = 0$. *Since we need to find* $\frac{1}{\det(\mathbf{A})} = \frac{1}{0}$, *we cannot divide by zero, so this matrix does not have a matrix.*

The reason happens when we use elementary row operations

$$\begin{pmatrix} 1 & -2 \\ -3 & 6 \end{pmatrix} \stackrel{3R_1 + R_2 \to R_1}{=} \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

This property of $\det(\mathbf{A}) = 0$ implies that $\mathbf{A}^{-1}$ does not exist works for all type of matrices.

1.2.2 Determinants and inverses of 3×3 matrices

Let

$$\mathbf{A} = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix},$$

Then the 3×3 matrix, has a determinant of

$$|\mathbf{A}| = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

While you can find the inverse of a matrix of any size based on the determinant and other linear algebra topics, we will use what we have already figured out to find the inverse of a 3×3 matrix.

One way to find the inverse of this matrix, $\mathbf{A}^{-1}$ is to augment $\mathbf{A}$ with the identity matrix, $\mathbf{I}_3$, meaning $\begin{pmatrix} \mathbf{A} & | & \mathbf{I}_3 \end{pmatrix}$. This would result in

$$\left(\begin{array}{ccc|ccc} 3 & 0 & 2 & 1 & 0 & 0 \\ 2 & 0 & -2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right).$$

Now we use elementary row operations take $\mathbf{A}$ to $\mathbf{I}$. And when you do this operation for the augmented matrix, you end up with $\left(\mathbf{I}_3 \quad \middle| \quad \mathbf{A}^{-1}\right)$. Using this technique, find $\mathbf{A}^{-1}$.

$$\left(\begin{array}{ccc|ccc} 3 & 0 & 2 & 1 & 0 & 0 \\ 2 & 0 & -2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \overset{R_1+R_2\to R_1}{=} \left(\begin{array}{ccc|ccc} 5 & 0 & 0 & 1 & 1 & 0 \\ 2 & 0 & -2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right)$$

$$\overset{0.2R_1\to R_1}{=} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0.2 & 0.2 & 0 \\ 2 & 0 & -2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right)$$

$$\overset{2R_1-R_2\to R_2}{=} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0.2 & 0.2 & 0 \\ 0 & 0 & -2 & -0.4 & 0.6 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right)$$

$$\overset{-0.5R_2\to R_2}{=} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0.2 & 0.2 & 0 \\ 0 & 0 & 1 & 0.2 & -0.3 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right)$$

$$\overset{R_2\leftrightarrow R_3}{=} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0.2 & 0.2 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0.2 & -0.3 & 0 \end{array}\right)$$

$$\overset{R_3-R_1\to R_3}{=} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0.2 & 0.2 & 0 \\ 0 & 1 & 0 & -0.2 & 0.3 & 1 \\ 0 & 0 & 1 & 0.2 & -0.3 & 0 \end{array}\right)$$